Aashish Pokhrel

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Website | GitHub | Google Scholar

Research Interests

Reliable multimodal intelligence, particularly language grounding, negation, and compositional reasoning in vision-language models. Additional interests include human-centered AI, efficient multimodal sensing for wearable and assistive systems, and 3D scene understanding.

Education

University of Wyoming Aug 2025 – May 2027 (Expected)
Master of Science, Computer Science; GPA: 4.0/4.0
Advisor: Dr. Shivanand Venkanna Sheshappanavar
Relevant Coursework: LLMs, Deep Learning, Explainable AI, Analysis of Algorithms

Institute of Engineering (IoE), Tribhuvan University Dec 2016 – Jan 2021
Bachelor of Engineering, Computer Engineering

Publications

1. **Aashish Pokhrel**, Bipin Ghimire, Prashanna Mani Paudel, Shivanand Venkanna Sheshappanavar. "Negation in Vision-Language Models: A Survey." *International Conference on Pattern Recognition (ICPR)*, 2026. [Paper] [Code]
2. **Aashish Pokhrel**, Shivanand Venkanna Sheshappanavar. "Negation Matters: Training-Free Negation-Aware Image Retrieval." *GRAIL-V Workshop, CVPR*, 2026. [Paper] [Code]
3. Bipin Ghimire, **Aashish Pokhrel**, Prashanna Mani Paudel, Shivanand Venkanna Sheshappanavar. "EgoThermal: An Egocentric RGB-D-T Dataset and Benchmark for Physical Awareness in Wearable Assistive Device." *EOF Workshop, WACV*, 2027 (accepted).

Research Experience

Geometric Intelligence Research Lab, University of Wyoming Aug 2025 – Present
Graduate Research Assistant, advised by Dr. Shivanand Venkanna Sheshappanavar

- **Post-hoc negation for frozen CLIP.** Developed a lightweight framework combining a text decomposer with fewer than 2M parameters and a single-parameter polarity correction. Improved multiple-choice negation accuracy from 74.3% to 79.8% on COCO and from 81.1% to 89.3% on VOC2007 against evaluated post-hoc baselines. Diagnosed limitations of rotation-based correction through bootstrap confidence intervals, per-object calibration, and angular zone analysis. Manuscript in preparation.
- **SpaceVLM-DRC.** Developed a training-free retrieval approach combining LLM-based query decomposition with dynamic embedding repulsion; achieved 53.0% R@5 on MSRVTT Retrieval-Neg.
- **Cultural heritage demo.** Built a web demo of negation-aware image retrieval over a large collection, running on the university HPC cluster.
- **EgoThermal.** Aligned depth and thermal camera streams for an egocentric RGB-D-T dataset and benchmark for wearable assistive devices, tested the pipeline on edge hardware (NVIDIA Jetson and Raspberry Pi), and annotated about 500 images. Accepted at the EOF Workshop, WACV 2027.
- **3D benchmarks.** Annotated approximately 800 samples for human evaluation of unsolvable-problem detection in 3D-LLMs (UPD-3D). Cleaned approximately 1,200 meshes in MeshLab for GroceryWild500, a novel-view-synthesis benchmark of grocery-store scenes.

Professional Experience

Ycotek AI Jan 2025 – Jul 2025

Senior Software Engineer I, Data | Clients: Torrens University, Media Design School, Strategic Education

- Analyzed data flows across 12+ enterprise systems to find integration gaps, then redesigned the pipelines, improving data integration accuracy and performance by 35%.
- Defined data-access governance requirements with stakeholders at 8 institutions and built Power BI models with row-level security, cutting client-reported data escalations by 90%.

Logic Information Systems (now acquired by Accenture) Jul 2021 – Dec 2024

Software Engineer, Data | Clients: Ralph Lauren, Belk, Samworth Brothers (UK), Blue Mercury, Express

- Built an automated sales-monitoring pipeline using anomaly detection, cutting analyst effort by 95% and speeding decisions by 40%.
- Automated associate-level sales scorecards for bonus decisions, cutting reporting turnaround by 60%.

Presentations & Talks

- "Negation in Vision-Language Models: A Survey," Poster, *International Conference on Pattern Recognition (ICPR)*, August 2026.
- "Negation Matters: Training-Free Negation-Aware Image Retrieval," Poster, *GRAIL-V Workshop, CVPR* 2026, and *SI Research Symposium*, University of Wyoming, 2026.
- "Saying No by Asking Yes: Affirmative Decomposition and Evidence Reflection for Frozen CLIP," Poster, *Wyoming Computing Symposium*, October 2026.
- "Negation Understanding in Vision-Language Models," Talk, *Mountain West AI & HPC Summit*, University of Wyoming, September 2026.

Honors, Awards & Funding

- Graduate Research Assistantship, University of Wyoming, 2025–2027.
- Full Scholarship, Bachelor of Engineering, Tribhuvan University, 2016. Ranked 412 of 10,000+ students (top 5%) in the IoE national entrance examination.

Technical Skills

Languages & AI Frameworks: Python, R, SQL, C++; PyTorch, TensorFlow, scikit-learn
Cloud & HPC: GCP, AWS (S3, Lambda), Microsoft Azure, PBS Pro / SLURM cluster computing
Data & BI (prior industry): Snowflake, BigQuery, PostgreSQL, PySpark, Airflow, dbt, Tableau, Power BI
Edge & Sensing: NVIDIA Jetson, Raspberry Pi, depth and thermal camera alignment
Tools: Git, LaTeX, MeshLab, Scrum/Agile